

PERFORMANCE WORK STATEMENT

1 **Scope:**

- 1.1 The current water processing system at Naval Station Norfolk (NSN) relies on an outdated Programmable Logic Controller (PLC) system that is nearing the end of its operational life. This system is critical for controlling and monitoring essential processes, including pumps, valves, level indications, and alarms. To improve maintenance efficiency and reduce operational complexity, the contractors will provide an upgraded system that is a direct duplicate of an existing system at a separate site Norfolk Naval Shipyard (NNSY), ensuring component and software interchangeability. This project will modernize the control system to enhance reliability, efficiency, and security while achieving standardization with existing infrastructure.

2 **Period of Performance:**

- 2.1 The period of performance for this contract is 12 months from the date of the award, with the option for a 6 month extension.

3 **Place of Performance:**

- 3.1 The contractor will bring all required tooling and equipment to accomplish all work at NSN, within building CEP-198 with the exception of prefabrication of processing equipment and software development.

4 **References:**

The following documents and their references will be used in the technical requirements for performing work under this contract. The most current reference revision will be in effect at the time of award, unless otherwise specified. All work will be performed in accordance with the drawings, documents and procedures cited below regardless of if the drawings, documents and procedures are vessel specific. The revision/change levels of any NAVSEA drawings or document utilized will be dictated by the most current revision/change unless directed otherwise via an Official Shipyard Document or other NAVSEA approved document for performing work under this Performance Work Statement (PWS). If the Contractor desires to use any drawing or technical instruction outside of these guidelines, TPOC concurrence must be given in writing prior to using any other reference, via a Contractor deficiency reporting process to include resolution. Any document not available to the Contractor can be obtained by request via the NNSY Contracting Officer (KO).

- 4.1 Code of Federal Regulation (CFR) Title 49;
- 4.2 Code of Federal Regulation, Title 10
- 4.3 Radiological Controls for Shipyards (NAVSEA 389-0288), Revision 21
- 4.4 Code of Federal Regulations - 29 CFR 1915, Title 29; Occupational Safety and Health Standards for Shipyard Employment
- 4.5 Code of Federal Regulations 40; Protection of Environment
- 4.6 Code of Federal Regulations 19; 1910; Occupational Safety and Health Standards
- 4.7 NAVSEA Standard Item (SI) 009-01; General Criteria; accomplish
- 4.8 NAVSEA Standard Item (SI) 009-02; Environmental Compliance Report for Material Usage; accomplish
- 4.9 NAVSEA Standard Item (SI) 009-03; Toxic and Hazardous Substance; control
- 4.10 NAVSEA Standard Item (SI) 009-04; Quality Management System; provide
- 4.11 NAVSEA Standard Item (SI) 009-06; Maintaining Protection and Cleanliness from Non-Radioactive Operation; accomplish
- 4.12 NAVSEA Standard Item (SI) 009-09 Process Control Procedure (PCP); provide and accomplish
- 4.13 NAVSEA Standard Item (SI) 009-60; Schedule and Associated Reports for Availabilities Over 9 Weeks in Duration; provide and manage
- 4.14 NAVSEA Standard Item (SI) 009-74; Occupational, Safety and Health Plan; accomplish
- 4.15 NAVSEA Standard Item (SI) 009-120; Fact Finding and Critique of Unplanned Event; manage
- 4.16 Local Standard Item 042-26-001 Security Requirements for Work Within Norfolk Naval Shipyard (NNSY); accomplish
- 4.17 Local Standard Item 042-11-001 General Occupational Safety, Health and Environment (OSHE) Requirements for Work Within Norfolk Naval Shipyard; accomplish (PCP)
- 4.18 OSHECM Chapter 250 'Hazardous Energy Control'
- 4.19 Norfolk Naval Shipyard (NNSY) Asset Authorization Submittal Requirements
- 4.20 NAVFAC P-307; Weight Handling Program Management
- 4.21 Local Standard Item 042-22-001 General Requirements for Work Within Norfolk Naval Shipyard (NNSY); accomplish
- 4.22 NAVMED P-5055 Radiation Health Protection Manual
- 4.23 PLC Ladder Logic for Pumps in NRMD.
- 4.24 NAVSEA DWG 2380-004: Electrical Isolation Schematic at NRMD.
- 4.25 DWG 2380-004: Electrical Isolation Schematic at NRMD.
- 4.26 NPM Chapter XIII-C-15:Naval Station Norfolk Controlled Industrial Facility (CIF) Radioactive Liquid Waste Processing System Operating Procedure.
- 4.27 DWG 2380-003: NRMD RLW Status Board.

5 **REQUIREMENTS:**

5.1 **NNSY Requirements:**

- 5.1.1 NNSY will provide water and power to the facility in which the processing equipment is being installed. Power and water will be provided to the facility by existing services.
- 5.1.2 NNSY will provide schematic of RLW processing system and electrical diagram of the RLW system for isolation and installation.
- 5.1.3 NNSY will support Contractor work on 1st shift (**1st shift hours: 0720 to 1550**). Monday through Friday, excluding government holidays. Notification to the COR is required to support work outside of normal business hours 0720 to 1550. The government will not be compensating for overtime hours. Notification must be received by the close of business on Tuesday of the week prior to the week additional support is needed. Support outside of normal business hours will be evaluated on a case-by-case basis.
- 5.1.4 NNSY will maintain a custodian work force to include inspections and small work items outside of the contractor's work areas, maintaining the fire alarms and alarm system and monitoring shipyard workers access control. Contractor is required to maintain contractor work force access control.
- 5.1.5 NNSY will maintain facility security in CEP-198, where the process controls are located.
- 5.1.6 NNSY will provide technical representative points of contact for interface. See 5.17.
- 5.1.7 Contractor is subject to radiological oversight by NNSY at all times per the Quality Assurance Surveillance Plan (QASP).
- 5.1.8 NNSY will provide sufficient personnel to escort the contractor while in CEP-198.
- 5.1.9 NNSY will perform all lifts requiring rigging inside of the CEP-198 high bay.

5.2 General Requirements: In the performance of the resulting contract, the contractor will:

5.2.1 The contractor will perform all work in accordance with all applicable references identified in section 4. The contractor will comply with references as cited within this PWS and the current contract modification, unless otherwise authorized by the Government. The effective issue of Government approved documents referenced in the ship specifications or otherwise invoked in this contract, including revisions or amendments, shall be the latest document date prior to contract award that has been accepted by the Contractor. The contractor will promptly notify the government of all revisions identified to the documents referenced in section 4 and identify all cost and schedule impacts. Any changes within scope should be authorized by KO; out of scope work will not be compensated by Govt. Individual process, system, and component specification upgrades will be addressed as identified in the Final Work Package. These requests will be evaluated prior to implementation based on their economy, feasibility and practicality. Implementation shall be in accordance with standard shipyard procedures identified in the Quality Assurance Program Requirements.

5.2.2 The awarded Contractor's project manager will personally notify at least one senior Naval Nuclear Propulsion Program (NNPP) representative, the COR, and the Contracting Officer of any issue related to licenses, permits, agreements, consents, or other legal authorization required to perform work specified herein that would adversely affect the Contractor's ability to perform the functions defined in the contract. This notification is required whether or not such documents are identified in this specification. This notification will be via teleconference within one (1) business day identification of the issue and shall be followed up in writing within five (5) calendar days. Additionally, senior Contractor personnel shall personally inform at least one senior NNPP representative and the COR prior to initiating any discussion, oral or written, with State, Federal, or local regulators concerning the handling, packaging, or transportation of NNPP waste. This notification will occur under all circumstances even if the Contractor does not specifically identify that the NNPP is the waste generator. This notification is in addition to, and not in lieu of, the Contractor's responsibility to promptly notify the Contracting Officer of any issues that could adversely affect schedule, impact performance or result in noncompliance with the contract. The senior Contractor Personnel and senior NNPP

representative(s) for this contract will be designated at time of contract award (and updated as necessary).

5.2.3 Contractor will follow all security requirements of reference (4.15) and Safety, Health, and Environment requirements of reference (4.16).

5.2.4 The contractor will provide to the COR an updated POA&M and schedule for all associated work. A POA&M shall include key events and milestones with associated dates with planning and execution.

5.2.5 The contractor will submit Contract Funds Status Reports (CFSR) and the status of work reports. CFSRs and status of work reports shall be provided to the Government for all subcontracts. Should any subcontractors in a fixed price arrangement be willing to provide similar data, it should be integrated in the Contractor's data as a part of the completion report. (CDRL A009).

5.2.6 The contractor will provide Project Management to participate in the following meetings and conferences:

5.2.6.1 The Cost Report meeting every 2 weeks beginning at 1 month before beginning work through completion of contracted work.

5.2.6.2 Weekly Progress meeting. 1 meeting per week beginning at 1 month before beginning work through completion of contracted work.

5.2.6.3 Periodic availability review conferences at 25%, 50%, 75% accomplishment of work.

5.2.6.4 Daily Production meetings with the COR/ACOR.

5.2.6.5 Both a Pre-Job brief and a Pre-Shift brief will be accomplished before execution of approved work. The Pre-Job brief will be led by the worksite supervisor and include the mechanics and any support personnel for this job. The Pre-Job brief will entail a review of the entire TGI/CFR being worked at any time before the execution of the work.

A Pre-Shift brief will entail the work to be accomplished per the approved TGI/CFR on the day of execution. The Pre-Shift brief will be led by the mechanic and should include participation/engagement from all personnel executing the approved work to ensure all personnel understand the scope.

5.2.6.6 Other meetings as required to cover any emergent needs that may develop over the availability.

5.2.7 The contractor will furnish the material, services, and work space/trailers, unless furnished by the Government under express provisions of this contract, necessary for the execution of work.

5.2.8 Contractor will be responsible for removal and disposal of components, including but not limited to complete disassembly of the Programmable Logic Controller cabinet, electrical buttons, switches, wiring, panels, Motor Control Center relays, input/output cards, Control Stations, Central Processing Unit and insulation. Facility infrastructure of work location will not be affected or disturbed. (CDRL A004).

5.2.9 Contractor will perform all planning for work to remove and properly disposal of all items (including potentially hazardous waste) as necessary to include: (CDRL A004)

Engineering Services

Project management and Production Services

5.2.10 Work Items, specifications and other work products will be completely and thoroughly checked and reviewed by the Contractor for technical accuracy and compliance with provisions of technical instructions, specifications and assignments. The Government is the ultimate approval authority for work item specifications provided by the Contractor. Data and other work products developed by the Contractor for Contractor's accomplished work will be submitted to the COR for review by NNSY for approval before beginning work on the work item submitted. NNSY will review/comment and approve all work items submitted by the Contractor. NNSY will expedite review on documents once contractor gives NNSY a priority list based on schedule. The contractor will report any corrections necessary due to error or omission. No work may proceed without written approval of NNSY.

- 5.2.10.1 All work performed by the contractor must be in accordance with written and approved (by NNSY) work packages. Contractor must bring approved paperwork (ex.TGI/CFR/WP) to the job site while executing work.
- 5.2.10.2 NNSY can issue a pause/stopping point to the contractor for questions about how the contractor is executing work to gain alignment.
- 5.2.11 The contractor will submit reports as specified throughout this PWS to the COR.
- 5.2.12 The contractor will comply with all laws, ordinances, and regulations of Federal, State, and Local authorities regarding licenses, permits and consents that may be necessary. For any supplies or services provided pursuant to this contract, the contractor will comply with all applicable laws and regulations, including, but not limited to, the Nuclear Regulatory Commission (NRC), the Environmental Protection Agency (EPA), the Occupational Safety and Health Act (OSHA) and associated regulations, the Department of Transportation (DOT), the Toxic Substances Control Act (TSCA) and associated regulations, radioactive waste compact, and the appropriate Agreement State (as applicable).
- 5.2.13 The contractor will allow inspections of the work area by Naval Nuclear Propulsion Program (NNPP) Representatives at all reasonable times.
- Inspections will include radiological control, safety, and adherence to approved work procedures. All deficiencies documented by these inspections will be provided to the COR. Inspections may be at random.
- 5.2.14 The Contractor, will not take possession, custody, or control of the facility, but will, to the extent legally permissible, assume all risk, risk of loss, and all other incidents of ownership to the contractor work area within the facility that is under contractor control. The contractor will be liable and responsible to the extent allowed by applicable laws and regulations in any action that may claim damages or compensation in any connection with the facility after the Contractor has started work on the system.

5.2.15 Contractor will obtain authorization for all Portable Electronic Devices (PEDs) from NNSY per reference (4.18).

5.2.15.1 Contractors are authorized to bring the following privately owned personal electronic devices (PEDs) onto the Shipyard and in the Controlled Industrial Area (CIA), but not within secured spaces: cell phones and Bluetooth PEDs, including hands-free (earpieces). It is required to leave these devices outside of secured spaces. Those secured spaces include Open Storage Areas, Nuclear Work Areas, and Controlled Nuclear Information Areas.

5.2.15.2 A PED is defined as any non-stationary electronic apparatus with singular or multiple capabilities of, but not limited to recording, storing, and/or transmitting data, voice, video, or photo images (e.g., cell phones, laptops, tablets and wearable devices such as fitness bands and smart watches).

5.2.15.2.1 NAVSEA 04 Ser 04/281 IMPLEMENTATION PLAN FOR ALLOWING CAMERA-PORTABLE ELECTRONIC DEVICES WITHIN THE CONTROLLED INDUSTRIAL AREA

5.2.15.2.2 NAVSEAINST 2200.1A PORTABLE ELECTRONIC DEVICES POLICY

5.2.15.2.3 SECROMAN Vol 1 5510.36 Rev E

5.2.15.3 In accordance with NAVSEA 04 Ser 04/281, personally owned camera enabled PEDs will be authorized inside of the Controlled Industrial Area (CIA) and all

5.2.15.4 Security Islands. Camera enabled PEDs will remain unauthorized within Controlled Nuclear Information Areas (CNIAs), Nuclear Work Areas (NWAs) and Open Storage Areas.

5.2.15.5 All PEDs, camera enabled or not, are still prohibited within NWAs and Open Storage Areas.

- 5.2.15.6 The allowance of camera enabled PEDs inside of the CIA and Security Islands does not negate existing photography policies. Unofficial photography, filming of video, or live streaming from the camera-enabled PED remains prohibited within the CIA or any NAVSEA controlled space outside of the CIA. Further, audio recordings (both digitally or magnetically) in NAVSEA spaces is prohibited without the approval of the Code 1120 Security Director.
- 5.2.15.7 Submarine SPECIFIC PEDs Instructions: In accordance with COMSUBLANT/ COMSUBPACINST 2075.1C, the use of PEDs while in or on Department of Navy Submarines are prohibited.
- 5.2.15.8 Camera cell phones are permitted on NSN; however, they cannot be brought into secured spaces where NAVSEA business is conducted.
- 5.2.15.9 PEDs brought within Shipyard worksites are subject to random audits. Unofficial photography is prohibited within the CIA and any Naval Sea Systems Command controlled space outside of the CIA. Photography and Videography is not permitted unless a permit is approved by the Command Security Manager, including video calls. It is required for the camera and photos taken to be submitted to security for review and approval. Devices found to contain government information shall be sanitized. The device will be examined to determine if government information can be removed without destruction. If it is determined that physical destruction is the only method for sanitizing, the device will be destroyed and the owner will not be reimbursed for the device.

5.2.16 REQUIREMENTS FOR ACCESS TO NAVY INSTALLATIONS

- 5.2.16.1 Commander, Navy Installations Command (CNIC), has established the Defense Biometric Identification System, (DBIDS), a standardized process for granting unescorted access privileges to vendors, contractors, suppliers and service providers not otherwise entitled to the issuance of a Common Access Card (CAC) who seek access to and can provide justification to enter Navy installations and facilities. Participation in the DBIDS is not mandatory, and if the Contractor chooses to not participate, the

Contractor's personnel will have to obtain daily passes, be subject to daily mandatory vehicle inspection, and will have limited access to the installation. The Government will not be responsible for any cost or lost time associated with obtaining daily passes or added vehicle inspections incurred by non-participants in the DBIDS. Alternatively, if the vendor so chooses, it may voluntarily elect to obtain long-term credentials through enrollment, registration, background vetting, screening, issuance of credentials, and electronic validation of credentials. Further information regarding DBIDS can be found at www.cnmc.navy.mil/om/dbids.html

NOTE: Bldg 276 TELEPHONE: 757-396-5815 (or 5095)

5.2.17 NORFOLK NAVAL SHIPYARD CONTRACTS DEPARTMENT
– SERVICES CONTRACTS DIVISION

- 5.2.17.1 No instructions, either verbal or written, deviating from the terms and conditions of this order are effective or binding upon the Government unless authorized and formalized by contractual documents issued by the Contracting Officer. On any problems concerning the order, please contact, CHAD GODWIN, CODE 411 NORFOLK NAVAL SHIPYARD CONTRACTS DEPARTMENT, SERVICE CONTRACTS DIVISION, BLDG. 65, 1ST FLOOR, PORTSMOUTH, VA 23709-5000.

5.3 Personnel Qualifications:

5.3.1 General:

5.3.1.1 All work under the terms of this contract will be performed in an efficient and workmanlike manner by personnel who are thoroughly familiar with the type of work being performed. The Contractor warrants that employees used on the work will, on the average, be as efficient as the average employees performing similar duties in the Contractor's own organization and will in all cases meet or exceed the minimum requirements set forth for the respective labor categories.

5.3.1.2 The contractor will be responsible for employing appropriate professionally and technically qualified personnel to perform the tasks outlined herein. The Contractor must have the qualified personnel, organization and administrative control necessary to ensure that the finished products shall conform to each task order. If the Contracting Officer questions the qualifications or competence of any person(s) performing the contract, the burden of proof to sustain the person(s) is (are) qualified as prescribed herein will be upon the Contractor.

5.3.1.3 The level of skill, education, and experience of the personnel required to perform this effort will meet the minimums specified below. Personnel assigned for performance of a task to be ordered under this contract shall meet the minimum qualifications for the respective labor category under which they are charged.

5.3.2 Minimum Qualifications:

5.3.2.1 Personnel assigned to or utilized by the Contractor in the performance of this contract shall, as a minimum, meet the experience, educational, or other background requirements set forth in Section 5.3.3 and be fully capable of performing the contemplated functions of the respective labor categories in an efficient, reliable, and professional manner.

5.3.2.2 The Government reserves the right to request the qualifications or competence of any person performing under this contract and the burden of proof to sustain that person is qualified as prescribed herein shall be upon the Contractor.

5.3.2.3 The contractor will be responsible for employing trained management, technical, and other personnel to perform work outlined. The Contractor must have the personnel, organization, and administrative control necessary to ensure that the services they perform meet all requirements specified in the delivery orders. The work history of each Contractor employee shall contain experiences directly related to the tasks and functions they are intended to perform prior to performing services under the contract. Further, these prior work experiences must be of sufficient variety and duration that the employee is able to effectively and efficiently perform the tasks and functions they will be assigned.

5.3.2.3.1 Training and qualification requirements apply to any subcontractors in the same manner with no exceptions.

5.3.3 Engineering requirements:

5.3.3.1 Academic Requirements: Must have a Professional Engineer's License, (as appropriate to assigned cognizance), or a Bachelor's degree in Engineering (as appropriate to assigned cognizance). A degree in Engineering Technology is not considered equivalent to the required Bachelor of Engineering Degree.

5.3.3.2 Specialized Experience Requirements (in area of assigned cognizance): A minimum of five (5) years practical design engineering experience, at a professional level, in responsible engineering duties, additionally a Minimum of three (3) years' experience working with Programmable Logic Controllers and Process Controls.

5.3.4 Contractor will provide NNSY with a list of all personnel working on this contract. (CDRL A001)

5.4 General Radiological Requirements:

- 5.4.1 The contractor will perform all work in accordance with reference 4.3. In all disputes in the interpretation of reference 4.3, NNSY is the final authority.
 - 5.4.1.1 Per reference (4.3) the following conditions are considered radiological incidents. If an incident occurs the contractor must immediately suspend all work and report it to NNSY. Contractors must then submit a recovery plan to the COR for approval. For convenience, section numbers of reference (4.3) are listed after each incident.
 - 5.4.1.2 External radiation exposure in excess of NAVSEA limits (210.1).
 - 5.4.1.3 Internally deposited radioactivity (410.2).
 - 5.4.1.4 Skin Contamination (510.1 and 510.2).
 - 5.4.1.5 Personnel exposed to airborne radioactivity (410.1).
 - 5.4.1.6 Unauthorized radioactive liquid discharge to the environment (310.1 and 310.2).
 - 5.4.1.7 Loss of radioactive material (710.1).
 - 5.4.1.8 Spread of radioactive contamination (510.3 and 510.4).
 - 5.4.1.9 Unmonitored exposure (210.2).
 - 5.4.1.10 Improper control of high radiation areas or exclusion areas (210.3).
 - 5.4.1.11 Unauthorized disposal of radioactive waste. Improper off-site disposal of mixed waste or PCB/radioactive waste (310.3).
 - 5.4.1.12 Improperly controlled radioactive material (710.2).
 - 5.4.1.13 Dosimetry discrepancy (210.5).

- 5.4.1.14 Any instance in which personnel radiation exposure exceeded the control levels of 106.1.a, 106.8.b, and 106.9 (210.6).
- 5.4.1.15 Loss, improper control, or leaking of radioactive sources (710.3).
- 5.4.2 Contractor personnel will wear dosimetry per Reference 4.3.
- 5.4.3 Contractor personnel will be entered into NNSYs dosimetry program and subject to the rules and regulations of the program. Personnel will be required to sign a form with the following statement: "I know of no medical condition that would prevent my receiving radiation exposure within prescribed Federal Standards, and attend a brief." NNSY will provide the brief. (Reference 4.21).
- 5.4.4 The contractor will provide the COR a Point of Contact (POC) listing citing all contractor personnel for Radiological Manager's notification to NNSY RADCON, to include the following: Radiological Control Department, Radiological Waste Management, Radiation Worker Training, and Radioactive Material Control Office. (CDRL A002)
- 5.4.5 When a problem occurs that meets or potentially could meet the criteria listed in 5.4.1.1 – 5.4.1.15 the contractor must stop work and perform a Team Learning Session (TLS) or equivalent. This TLS includes a discussion (and written report) of what happened, what should have happened, what the team learned, and what actions are going to be taken to ensure that the problem will not happen again. TLS must include the COR and any technical representatives that the COR deems necessary.
 - 5.4.5.1 For problems that occur but the COR does not deem significant enough to have violated the criteria of 5.4.1.1 – 5.4.1.15, then the COR may only require a written report and not a full meeting.

5.5 Packaging of Materials by the Contractor:

- 5.5.1 Material must be properly packaged, blocked and braced per local, state, federal and DOT regulations.

5.6 System Assessment and Design (CDRL A003):

- 5.6.1 Conduct a comprehensive assessment of the existing PLC system to be upgraded.
- 5.6.2 Perform a detailed analysis of the separate, existing PLC system at NNSY that will serve as the template for this project to ensure the new system is an exact duplicate.
- 5.6.3 Develop a detailed drawing of the new system that guarantees it is identical to the design provided in Attachment 1. This drawing must be approved by the Government.

5.7 Hardware and Software Procurement:

- 5.7.1 Procure all necessary hardware (PLCs, HMIs, servers, network switches, flowmeters etc.) and software that is **identical and interchangeable** with the designated template system, to the maximum extent practicable. Any proposed deviation due to obsolescence must be approved in writing by the government. (CDRL A006).

5.8 System Development and Implementation:

- 5.8.1 Develop and configure the new PLC and SCADA/HMI applications to be identical to the template system.
- 5.8.2 Maximize the preparation of all hardware and software at contractor site to minimize installation time required on NNSY site.
- 5.8.3 Install and configure all new hardware and software. (CDRL A005)
- 5.8.4 Develop control logic and alarm settings for the new system that is functionally identical to that of the template system.
- 5.8.5 Integrate the new system with existing instrumentation and field devices.
- 5.8.6 Replace existing flowmeters with new flowmeters that are identical to those installed on the template system.

5.9 Electrical Standards:

- 5.9.1 The system design must adhere to relevant electrical standards, including: IEEE C37.1: IEEE Standard for SCADA and Automation Systems and ISA/IEC 62443: Security for industrial automation and control systems.
- 5.9.2 All new electrical installations must comply with the National Electrical Code (NEC)

5.10 Hazardous Energy Controls:

- 5.10.1 "Per FAR 22.102-1(a) entitled "Safety", Contractors are hereby informed that their employees or representatives present on property under Norfolk Naval Shipyard cognizance must follow Federal and State OSHA requirements and Norfolk Naval Shipyard OSHE Program Manual, NAVSHIPYDNORINST P5090.2 requirements or safety requirements for site specific location of the contract. Contractors and Non-NNSY Government Agencies (NNGAs) are hereby informed that their employees or representatives present on property under Norfolk Naval Shipyard (NNSY) cognizance must adhere to Federal OSHA requirements for hazardous energy control IAW 29 CFR 1915.89 and must follow Hazardous Energy Control (HEC) procedures as identified in NAVSEA's Occupational Safety, Health, and Environment Corporate Manual (OSHECM) Chapter 250 (latest revision), when performing work under NNSY contract. All HEC work to be performed by Contractors and NNGAs requires that NNSY Technical Points of Contact (TPOCs) and/or Contracting Officer's Representatives (CORs) be notified of the HEC scope of work prior to starting work. Any changes or deviations in the initial scope of HEC work requires the NNSY TPOCs and/or CORs be informed of the change/deviation. Appendix 250.B-1 through B-3 of OSHECM Chapter 250 for multiple employer work identifies when Contractors and NNGAs are required to assign their Lockout/Tags-Plus Coordinator (LOTC) to coordinate HEC work with the cognizant NNSY LOTC. HEC locks shall be provided by the contractor or NNGA for their employees use. The HEC locks shall have red bodies with name labels or name tags affixed, identifying who installed the HEC lock(s) and a point of contact telephone number. HEC lock name labels and tags must be made of materials that will hold up to the environment to which they are exposed. HEC tags used for Tags-Plus HEC work shall be provided by the cognizant NNSY LOTC to the contractor or NNGA for their

use if machinery, equipment, or systems (MES) cannot be locked out. OSHECM Chapter 250 Tags-Plus procedures will be strictly followed.

- 5.10.2 Contractors and NNGA's may obtain electronic copies of OSHECM Chapter 250 (latest revision) and access the OSHA 1915.89 web site by using the following link to NNSY's public access web site: <http://www.navsea.navy.mil/shipyards/norfolk/default.aspx>

Names and contact information for NNSY LOTCs and/or the Facility Custodian will be provided by NNSY TPOC and/or CORs set forth elsewhere in this award document."

5.11 Testing and Commissioning:

- 5.11.1 Develop and assist NNSY in executing a comprehensive test plan to verify that the new system meets all functional requirements and operates identically to the template system.

- 5.11.1.1 Contractor will NOT be authorized to operate the RLW system independently. All operations of the RLW system require NNSY approval and support.

- 5.11.2 Conduct a full system commissioning to ensure safe and efficient operation.

5.12 Documentation:

- 5.12.1 Provide a complete set of documentation for the new system, including:

- 5.12.1.1 A detailed map of all physical Inputs and Outputs (I/O), including the associated variable names within the ladder logic.

- 5.12.1.2 A step-by-step procedural document detailing the methodology for manually operating the pumps, solenoid valves and other essential equipment in the event of a total PLC system failure.

- 5.12.1.3 As built drawings, O&M manuals, and software configurations.

5.13 Training:

- 5.13.1 Contractor will provide a comprehensive 3-day, on-site training course for up to 12 personnel. This training shall focus on the practical use of the software interface for monitoring, troubleshooting, and system operation.

5.14 Laptop Provision

- 5.14.1 Contractor will provide one (1) new laptop computer, pre-loaded and fully configured with all necessary software to interface with both the newly upgraded PLC system and the existing template system. The laptop must meet or exceed specifications approved by the government. (CDRL A007)

5.15 Required Standard of Workmanship:

- 5.15.1 Unless otherwise specifically provided in the contract, the quality of all services rendered hereunder will conform to the highest standards in the relevant profession, trade or field of endeavor. All services will be rendered by or supervised directly by individuals fully qualified in the relevant profession, trade or field, and holding any licenses required by law and regulation.
- 5.15.2 For the purpose of the clause, the “highest standards” will be the generally prevailing standards in the process controls industry. In the event an industry-recognized authority subsequently issues a generally accepted standard for the industry, Contractor’s services will, as a minimum, meet such a standard.

5.16 Prior Written Permission Required for All Subcontracts:

- 5.16.1 Any changes in subcontractors additions of subcontractors, or change in the processes performed by subcontractors must be approved by the Contracting Officer prior to the change or addition. A Contractor's request to change or add subcontractors or subcontracted processes will include the Technical Capability Information from the solicitation, which is delineated in the 52.212-1 Instructions to Offerors provision of the solicitation, about the subcontractor(s) covered by the request. (CDRL A008).

5.17 Points of Contact : TBD

5.18 Post Award Actions

5.18.1 Contractor will submit a Safety plan to NNSY for review and approval prior to the commencement of work.

5.18.2 Contractor will enter into a Memorandum of Agreement with NNSY for lifts requiring rigging within the CEP-198 high bay.